

Template-Guided Cascaded Diffusion for Stylized Handwritten Chinese Text-Line Generation

Honglie Wang[1,2], Minsi Ren[1,2], Yan-Ming Zhang[1,2(✉)], Fei Yin[1,2], and Cheng-Lin Liu[1,2]

[1] State Key Laboratory of Multimodal Artificial Intelligence Systems (MAIS), Institute of Automation, Chinese Academy of Sciences, Beijing 100190, China
{wanghonglie2023,renminsi2021}@ia.ac.cn, {ymzhang,fyin,liucl}@nlpr.ia.ac.cn

[2] School of Artificial Intelligence, University of Chinese Academy of Sciences, Beijing 100049, China

Abstract. Current methods for handwritten Chinese text generation primarily focus on generating individual characters. However, for practical applications, users often require a handwritten text-line generation model to streamline their workflows. Generating stylized handwritten text-line necessitates capturing both character-level styles (e.g., stroke thickness, ink color, and cursive join) and text-line-level styles (e.g., inter-character spacing and text-line inclination). To address this challenge, we propose a novel cascaded diffusion architecture (i.e., Layout-Diff and Imitating-Diff) to disentangle these two levels of styles via customized template guidance. Specifically, Layout-Diff assists in synthesizing a template image, where the visual archetypes, rendered in a printed font, are well-arranged on a blank canvas. Subsequently, Imitating-Diff transforms this synthesized template into a stylized handwritten text-line image. We propose a latent aggregation module to efficiently incorporate textual guidance into Imitating-Diff while reducing computational cost. Furthermore, we design a high-frequency weighted loss for fine-tuning Imitating-Diff in the pixel space to preserve structure details with clarity. Extensive experiments demonstrate that our method effectively mimics diverse handwriting styles while ensuring structural accuracy in generated text lines.

Keywords: handwritten text-line generation · style disentanglement · diffusion model

1 Introduction

Handwritten text generation lies at the intersection of NLP and computer vision, aiming to synthesize legible and structurally accurate text that preserves individual writing styles and natural variability [18,44]. This capability underpins

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
X.-C. Yin et al. (Eds.): ICDAR 2025, LNCS 16023, pp. 149–166, 2026.
https://doi.org/10.1007/978-3-032-04614-7_9



Fig. 1. Several examples demonstrating the high-quality performance of our handwritten text-line generation model. The proposed architecture not only seamlessly emulates users' writing styles but also preserves the clarity and integrity of character structures.

applications such as personalized signatures, stylized emails, custom document rendering, and historical manuscript restoration [13].

In particular, Chinese character generation remains challenging due to the script's complexity and diversity [23,24,32,45,51]. While early methods were limited to fixed writer IDs [14], recent advances leverage one-shot learning to imitate styles from a single exemplar [5,47]. However, most existing approaches operate at the character level, failing to address line-level handwriting synthesis. This task requires resolving two key issues: (1) modeling inter-character spatial layout and stylistic attributes such as spacing and line inclination, and (2) preserving consistent character-level traits—e.g., stroke width, ink color, and ligatures—without compromising structural fidelity.

To address these challenges, a straightforward approach is to generate characters independently using a single-character model and manually stitch them according to a text-line layout (see Fig. 1). However, this method often produces rigid and unnatural text-lines, as it fails to capture the stylistic cohesion inherent in natural handwriting. Alternatively, directly constructing one generative model is typically challenging and impractical for precise editing, as it lacks an explicit representation of layout information. To better capture the spatial stylistic characteristics of text lines, we extend the concept of visual archetypes [31] from the character level to the text-line level, a construct we term **Template**. Template is synthesized by arranging resized standard character images from a printed font according to the target layout, thereby disentangling character-level styles and text-line-level styles to create stylized handwritten text lines. Com-

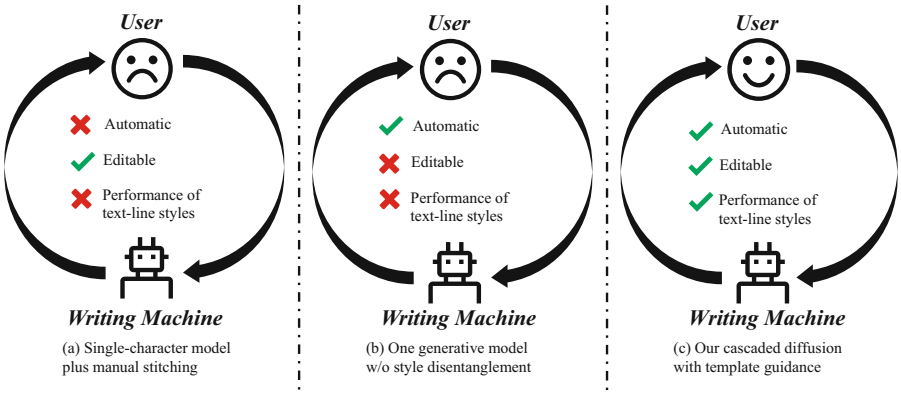


Fig. 2. User experience comparisons between three different methods. Our approach enables the generation of complete text-line images in a single process, seamless editing using a template and effective preservation of text-line styles.

pared to post-processing generated text-line images, directly editing the template offers greater flexibility for users and enhances convenience. To construct a high-quality template, we propose Layout-Diff, a lightweight 1D model that captures the character sizes and inter-character positional relationships within the text line.

To adapt a fixed-style template to diverse writing styles, we propose the Imitating-Diff model(see Fig. 2). This model ensures that each character 1) is positioned in alignment with the template, maintaining a similar size, 2) exhibits a consistent style that resembles the reference image, and 3) preserves both structural accuracy and readability. To reduce computational overhead, we utilize a pretrained and frozen VAE model from SD1.5 [28] to extract template features, ensuring that both noise and template condition features reside in the same latent space. To further exploit the high-frequency structural details of each character, we fine-tune Imitating-Diff in the pixel space with a frozen VAE decoder and high-frequency feature loss, which significantly enhances the readability of characters within the text-line image. Moreover, given the extreme aspect ratios of text-line images and limited computational resources, it is also crucial to train the model within the latent space.

Our contributions can be summarized as follows:

- We propose a novel cascaded diffusion framework designed to disentangle character-level and text-line-level styles. This framework comprises two models: Layout-Diff, which learns spatial styles, and Imitating-Diff, which mimics handwritten styles. These models are systematically integrated through a Template construct.
- We propose a latent aggregation module that integrates content as well as calligraphy style information into Imitating-Diff, effectively preserving the spatial layout and structural coherence of characters within the text-line while imitating the target writing style.

- We refine Imitating-Diff in the pixel space by incorporating a L1 loss weighted by a high-frequency mask, which enhances the clarity of fine-grained character details, thereby improving overall text-line readability.
- Extensive experiments on Chinese handwritten text-line datasets validate the effectiveness of our method, demonstrating robust performance in generating stylized handwriting at the line level.

2 Related Work

Handwritten data can be stored in the form of online trajectory sequences or offline images. The advantage of online handwritten data lies in its ability to maintain dynamic trajectory information of the drawing process, while the advantage of offline handwritten data lies in its more natural representation of visual features such as stroke thickness and ink color.

Online Handwriting Generation. Most of online handwriting generation methods are typically based on autoregressive neural network [12,49], utilizing a mixture of Gaussian distributions to model the movement of pen. SketchRNN [15], which based on Variational Autoencoders (VAE) [20], adopts RNN to conduct the task of unconditional sequence data generation. Subsequently, a series of works [1,22,40–42,50] focuses on conditional generation, aiming to produce handwriting that encompasses specific content or diverse writing styles. SDT [6] leverages the style-disentangled Transformer to more effectively extract style features from reference samples. In addition, there are studies that apply diffusion models to online sequential data [7,29,34], achieving notable results.

Offline Handwriting Generation. Offline handwriting generation methods [2,4,11,21,43] were primarily based on Generative Adversarial Networks (GANs) in the early days, leveraging adversarial training to create realistic handwriting samples. Recently, with the development of diffusion models, an increasing number of studies have begun to utilize these models to generate offline font data [46]. Diffusion models offer a novel framework that refines random noise into structured handwriting, allowing for more nuanced control over style and content. State-of-the-art methods can now extract style information from a significantly limited number of reference samples [5,9,30,47,52].

Score-Based Diffusion Model. Score-based diffusion models systematically perturb data via a forward diffusion process, subsequently reversing this process to recover the original data [17,36,38]. Due to their capability to balance fidelity and diversity, along with advancements in accelerated sampling [3,27,37] and conditional generation [8,35], these models have been extensively adopted across diverse data modalities. We also adopt diffusion models as our generative framework.

3 Preliminary

Diffusion Formulation. Following [39], we formulate the forward diffusion process and reverse denoising process as stochastic differential equations (SDEs). The forward SDE transforms the true data vectors $\mathbf{x}_0$ into Gaussian noise:

$$d\mathbf{x}_t = -\frac{1}{2}\beta_t \mathbf{x}_t dt + \sqrt{\beta_t} dw_t, \quad t \in [0, T], \tag{1}$$

where w_t represents standard Brownian motion, $t \in [0, T]$ with T as a finite time horizon, and β_t is a non-negative noise schedule function. The solution to this equation is given by:

$$\mathbf{x}_t = e^{-\frac{1}{2}\int_0^t \beta_s ds}\mathbf{x}_0 + \int_0^t \sqrt{\beta_s} e^{-\frac{1}{2}\int_0^t \beta_u du} dw_s. \tag{2}$$

By utilizing properties of Ito's integral, the conditional distribution of $\mathbf{x}_t$ given $\mathbf{x}_0$ is Gaussian:

$$p(\mathbf{x}_t|\mathbf{x}_0) \sim \mathcal{N}(\rho(\mathbf{x}_0, t), \lambda_t), \tag{3}$$

where $\rho(\mathbf{x}_0, t) = e^{-\frac{1}{2}\int_0^t \beta_s ds}\mathbf{x}_0$ and $\lambda_t = I - e^{-\int_0^t \beta_s ds}$. The reverse SDE transforms the Gaussian noise back into the data $\mathbf{x}_0$ using the following process:

$$d\mathbf{x}_t = -\left(\frac{1}{2}\mathbf{x}_t + \nabla \log p_t(\mathbf{x}_t)\right)\beta_t dt + \sqrt{\beta_t} d\tilde{w}_t, \quad t \in [0, T], \tag{4}$$

where $\tilde{w}$ denotes the reverse-time Brownian motion. Additionally, we can represent the reverse process as an ordinary differential equation (ODE):

$$dx_t = -\frac{1}{2}\left(\mathbf{x}_t + \nabla \log p_t(\mathbf{x}_t)\right)\beta_t dt, \quad t \in [0, T]. \tag{5}$$

We train the network as denoiser $\mathcal{E}_\theta$ to predict the original clean data from noisy data:

$$\hat{\mathbf{x}}_0 = \mathcal{E}_\theta(\mathbf{x}_t, t, c),$$

where c represents the additional conditions. We use the predicted $\mathbf{x}_t = 0$ to estimate the score $\nabla \log p_t(\mathbf{x}_t)$ using $\lambda_t^{-1}(\rho(\hat{x}_0, t) - x_t)$, allowing us to sample data $\mathbf{x}_0$ by starting from Gaussian noise $\mathbf{x}_T \sim \mathcal{N}(0, 1)$ and numerically solving the SDE in Equation (4) or the ODE in Equation (5). The loss function used to train the diffusion model is expressed as follows:

$$\ell_{\text{diff}} = \mathbb{E}_{\mathbf{x}_0, t}[\|\hat{\mathbf{x}}_0 - \mathbf{x}_0\|_2^2]. \tag{6}$$

4 Methods

4.1 Overview

Our goal is to synthesize handwritten text-line images given an arbitrary length string $\mathcal{A}$ as content and a single style reference image $\mathcal{I}_{style}$ from a writer w_s.

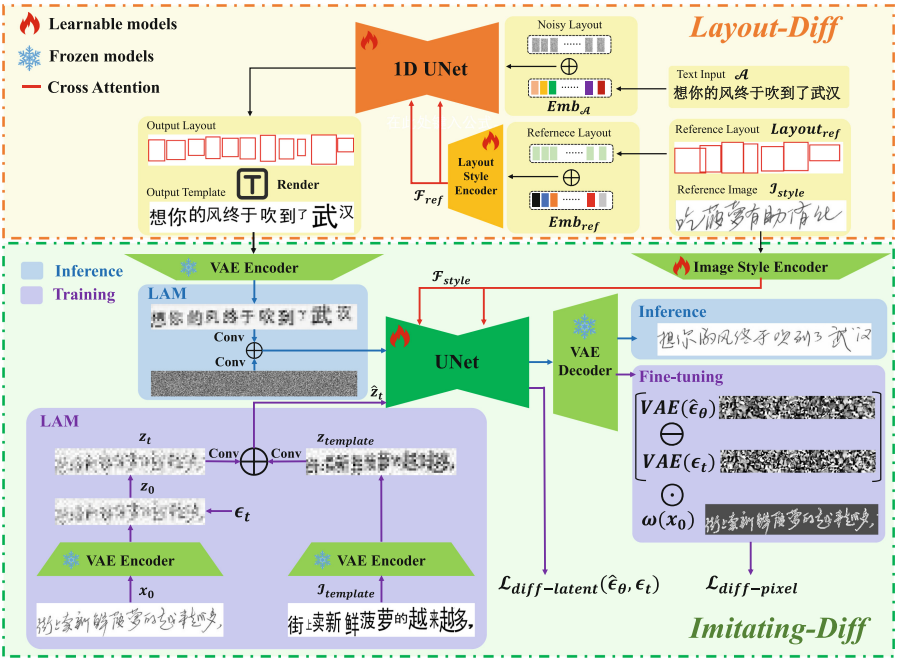


Fig. 3. The overall architecture of the proposed cascaded diffusion model. Given the user-provided text input $\mathcal{A}$ and style reference image $\mathcal{I}_{style}$, our framework is capable of generating customized handwritten text-line images. All vae modules remain frozen. LAM represents latent aggregation module.

Here, $\mathcal{A} = \{a_i\}_{i=1}^{L}$ is a sequence of length L, where each a_i is a Unicode character, encompassing Chinese characters, English letters, digits, punctuation and other symbols. The generated handwritten text-line images should *faithfully capture the distinctive style of* w_s *while accurately conveying the content of* $\mathcal{A}$. Furthermore, the spatial arrangement of characters must be fluid, maintaining the natural rhythm of handwritten text-line.

Existing methods *are limited to generating single Chinese characters* images and fail to handle more complex text content. To address this limitation, we propose a cascaded diffusion framework that systematically structures the generation process(see Fig. 3). Specifically, a layout style encoder and an image style encoder are employed to capture text-line-level and character-level styles, respectively. These multi-level style features are then injected into the corresponding UNet model via cross-attention. In Imitating-Diff, template information is fused with the noisy latent through concatenation. During training, the template is directly extracted from the input image, whereas during inference, it is synthesized by Layout-Diff.

4.2 Layout-Diff

Text-Line Template. As stated before, the objective of Layout-Diff is to generate a text-line template, which serves as the content information for Imitating-Diff. Text-line template comprises *the bounding box of each character* and their corresponding *standard font images*. Specifically, the bounding box defines the position and size of each character within the text-line image. The bounding box coordinates are converted into normalized values representing the height, width, the vertical distance from the top of each character to the top of the text line, and the horizontal distance between the centers of adjacent chactarers. We use the SimHei font as standard character images. We resize the height and width of standard character images and embed them into their corresponding bounding boxes in a blank canvas to construct the standard text-line-level templates.

Module Architecture. The layout of a text line refers to the bounding box coordinates for all the characters. As shown in Fig. 3, the Layout-Diff firstly takes the text input $\mathcal{A}$ and the layout of the style reference image $\mathcal{I}_{style}$ as input, and predicts the layout of the target text-line image. The generative process is modeled using the diffusion process, with a 1D U-Net network serving as the denoising model $\hat{\epsilon}_\theta$.

More concretely, considering that different characters have different shapes, (especially punctuation marks), we set a learnable encoding for each character class i: $Emb_i \in \mathbb{R}^d$, where d is the dimension of the encoding vector. To account for the distinct layout styles exhibited by different authors, we condition the model on the layout of the reference image $\mathcal{I}_{style}$. Specifically, the reference layout information $Layout_{ref}$, added with the reference character embeddings Emb_{ref}, is processed through a 1D layout style encoder $\mathbf{E}_{ref}$, which mirrors the architecture of the 1D U-Net encoder. The resulting style layout features $\mathcal{F}_{ref}$ are then fed into the denoiser via cross-attention.

Training Objective. We model the reconstruction loss as the average distance between the predicted layout and the actual counterpart. Here $\epsilon_t, \hat{\epsilon}_\theta$ represents the added true noise and predicted noise at timestep t, while $\mathbf{x}_t$ represents the noisy data at timestep t:

$$\begin{aligned} \mathcal{F}_{ref} &= \mathbf{E}_{ref}(Layout_{ref}, Emb_{ref}), \\ \mathcal{L}_{\text{diff-layout}} &= \mathbb{E}_{\epsilon,t}[\|\hat{\epsilon}_\theta(\mathbf{x}_t, t, Emb_{\mathcal{A}}, \mathcal{F}_{ref}) - \epsilon_t\|_2^2]. \end{aligned} \tag{7}$$

4.3 Imitating-Diff

The objective of Imitating-Diff is to transfer the text-line template image obtained to a stylized handwritten text-line image. A primary challenge lies in the significant computational cost, considering the enormous size of the text-line image. To address this, as shown in Fig. 3, our method integrates a pretrained frozen VAE, a CNN-Transformer style encoder, a latent aggregation module, and a UNet-based denoiser to form a **conditional latent diffusion model**.

Leveraging the pretrained VAE encoder, we reduce the resolution of all text-line images and templates by a factor of 8, transforming them from 96×2048 pixels to a 12×256 latent vectors. This reduction significantly saves computational resources, enhancing both training and inference efficiency. The generated latent features are finally mapped into the pixel space by the VAE decoder.

Another challenge is to ensure the accurate positioning and structure of characters, as defined in the template. For single-character generation models like One-DM [5], directly replacing the character template with the text-line template often results in improper allocation of characters, despite the presence of a content-style aggregation module. To address this issue, we design the Latent Aggregation Module to enable the model to better incorporate both content and style information.

Latent Aggregation Module. In the field of handwritten text generation, latent space-based diffusion methods [5,30,47,52] integrate text information into the diffusion model via cross-attention. In contrast, pixel space-based diffusion approaches, such as CG-DDPM [47], take a noisy image combined with a rendered glyph image as input to the U-Net. A key innovation of our work lies in demonstrating the effectiveness of template concatenation in the latent space, thereby eliminating the computational overhead associated with training a content encoder and employing additional cross-attention mechanisms.

As shown in Fig. 3, we compress the template image $\mathcal{I}_{template}$ into the latent template $\mathbf{z}_{template}$ using the same pretrained, frozen VAE encoder as latent diffusion model. Both the latent template $\mathbf{z}_{template}$ and the latent noise $\mathbf{z}_t$ are processed through separate convolutional layers, after which they are concatenated to $\hat{\mathbf{z}}_t$ and fed into the U-Net model for further computation. The latent template image, with its blurred spatial information, helps guide the layout during the entire generation process.

$$\begin{aligned} \mathbf{z}_{template} &= \mathbf{VAE}_{en}(\mathcal{I}_{template}), \\ \hat{\mathbf{z}}_t &= \mathbf{Conv}(\mathbf{z}_{template}) \oplus \mathbf{Conv}(\mathbf{z}_t). \end{aligned} \tag{8}$$

In order to mimic a specific writing style, the style image $\mathcal{I}_{style}$ is passed through a CNN-Transformer encoder $\mathbf{E}_{style}$ to extract the style features $\mathcal{F}_{style}$, which are injected into the denoiser backbone via cross-attention.

Training Objective. The training loss is defined in the latent space as the expectation of the $L2$ noise prediction error, denoted as $\mathcal{L}_{\text{diff-Latent}}$:

$$\begin{aligned} \mathcal{F}_{style} &= \mathbf{E}_{style}(\mathcal{I}_{style}), \\ \mathcal{L}_{\text{diff-latent}} &= \mathbb{E}_{\epsilon,t}[\|\hat{\epsilon}_\theta(\hat{\mathbf{z}}_t, t, \mathcal{F}_{style}) - \epsilon_t\|_2^2], \end{aligned} \tag{9}$$

where $\hat{\epsilon}_\theta$ represents the predicted noise modeled by Imitating-Diff.

Fine-Tuning in the Pixel Space. To address issues such as detail loss and stroke blurring caused by latent space diffusion models, we propose to fine-tune the Imitating-Diff in pixel space. The VAE decoder up-scales the latent

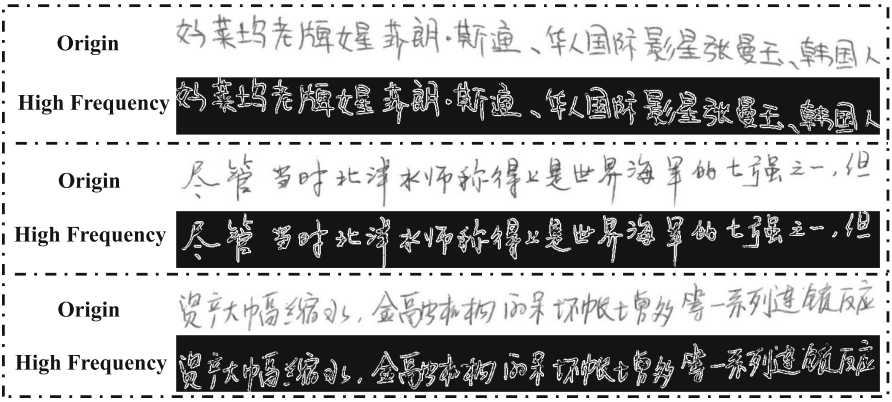


Fig. 4. Handwritten text-line images and corresponding high-frequency masks. We observe that character-level styles—such as stroke thickness, character contours, and cursive joins—are further enhanced and more prominently emphasized.

representations back to the original pixel space, allowing for the computation of the $L1$ distance in pixel space (see Fig. 3).

Moreover, to encourage the Imitating-Diff to focus on structural details and generate characters with clearer and smoother strokes, we employ a Sobel kernel to extract the high-frequency components of the target text-line image $\mathbf{x}_0$ to construct a high-frequency mask $Sobel(\mathbf{x}_0)$ with the same shape of $\mathbf{x}_0$ (see Fig. 4). The pixel-level loss $\mathcal{L}_{\text{diff-pixel}}$, is defined as the expectation of $L1$ noise prediction error weighted by high-frequency mask $\omega(\mathbf{x}_0)$. The final fine-tuning loss $\mathcal{L}_{\text{diff-ft}}$, is a weighted sum of $\mathcal{L}_{\text{diff-pixel}}$ and $\mathcal{L}_{\text{diff-latent}}$:

$$\begin{aligned} \mathcal{L}_{\text{diff-ft}} &= \lambda_{latent} \cdot \mathcal{L}_{\text{diff-latent}} + \mathcal{L}_{\text{diff-pixel}}, \\ \mathcal{L}_{\text{diff-pixel}} &= \mathbb{E}_{\epsilon,t}[\|\omega(\mathbf{x}_0) \odot (\mathbf{VAE}_{de}(\hat{\epsilon}_t) - \mathbf{VAE}_{de}(\epsilon_t))\|_1], \\ \omega(\mathbf{x}_0) &= \lambda_{Sobel} \cdot Sobel(\mathbf{x}_0) + \lambda_{all} \cdot \mathbf{1}, \end{aligned} \tag{10}$$

where $\mathbf{1}$ represents a matrix with the size of $\mathbf{x}_0$ in which all entries are equal to one. Notably, the pretrained VAE model parameters are kept frozen during this process.

5 Experiments

5.1 Experimental Settings

Evaluation Dataset. To evaluate the performance of our method, we use the CASIA-HWDB2.0-2.2 dataset [25] for training, which contains 52,230 Chinese text-lines from 1,019 distinct writers. For testing the style transfer capability of our method, we employ the ICDAR2013 dataset [48], which consists of 10,449 Chinese text-lines from 60 different writers. For layout generation, we set the

number of bounding boxes in a text line to 32, reflecting the typical writing habits of most individuals. Specifically, we pad target layouts with zeros and repeat the previous part of the text line for style layouts. For text-line image generation, all authentic text-line images are resized to a height of 96 pixels while maintaining their aspect ratio, and padded to a width of 2048 pixels. For template and target text-line images, we pad with pixels of value 255, while for style reference images, we repeat the previous part.

Evaluation Metrics. We evaluate the quality of layout generation based on multiple factors, including character height, width, vertical spacing, and the inter-character center distances. The L1 losses for all layout-related metrics are computed on a normalized image, where the height is scaled to 1.

To assess the style quality of text-line generation, we adopt three style metrics: the Fréchet Inception Distance (FID) [16], the Geometry Score (GS) [19] and the Style Score. For the style score, we train a CNN-Transformer encoder on the test dataset for style classification task. For all style metrics, we first compute the values for each writer in the test dataset, then calculate the mean across all writers' values.

To evaluate the structural accuracy of generated text-lines, we employ two content metrics: Correct Rate (CR) and Accuracy Rate (AR) [48]. These metrics are computed using a pretrained handwritten Chinese text line recognition model, which transcribes each text line into a character string.

Implementation Details. In all experiments, we use *only one style reference sample* to perform one-shot experiments (one text-line for layout and text-line generation). We train all conditional diffusion models on eight A6000 GPUs. Specifically, we train the Layout-Diff for 60,000 steps with a batch size of 1024. The Imitating-Diff is then trained for 200 epochs with a batch size of 20, followed by fine-tuning for 30 epochs with a batch size of 6. The fine-tuning loss weight ratio, λ_{latent} is 0.1, λ_{Sobel} is 0.7 and λ_{all} is 0.3. AdamW optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$ is used. The learning rate is set as 1e$-$4 with linear schedule for both training and fine-tuning. During inference, DDIM is used to speed up sampling with 25 steps.

Compared Methods. We compare our Imitating-Diff with one-shot handwritten character generation models, including FontDiffuser [46], GAC-Diff [47], and One-DM [5], all of which are adapted to generate coherent text-line images. For FontDiffuser and GAC-Diff, which already incorporate multi-scale content aggregation modules (or their variants), we simply replace the character template with a text-line template and substitute the font style reference with a text-line style reference. In the case of One-DM, we introduce additional positional embeddings (PE) or employ our proposed latent aggregation module (LA) to generate text-line images.

5.2 Evaluations

Text-line Layout Generation. We compare our Layout-Diff with other layout generation models [33] LayoutLSTM and LayoutTransformer. Quantitative

Table 1. The comparison of different layout generative methods.

	Height Loss	Width Loss	Vertical Loss	Horizontal Loss
LayoutLSTM	0.1133	0.1023	0.1137	0.1268
LayoutTransformer	0.1075	0.1067	0.1193	0.1222
LayoutDiffuser	**0.1028**	**0.0915**	**0.0673**	**0.1118**

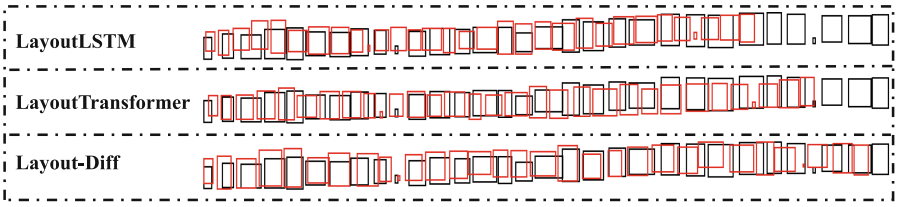


Fig. 5. Visualization of layouts generated by different methods. Black boxes denote the ground truth, while red boxes represent the generated layouts. The results illustrate that our layout generation model more effectively captures text-line-level style characteristics, demonstrating improved fidelity to the target structure. (Color figure online)

results are shown in Table 1. Through this in-context learning approach, the model can imitate the text-line-level style (e.g. textline slant) of the reference sample to a certain degree. However, Fig. 5 shows that both LayoutTransformer and LayoutLSTM underperform compared to the LayoutDiffuser model in terms of generating accurate character spacing and character size. We hypothesize that this underperformance may be attributed to exposure bias inherent in autoregressive models during the inference process, leading to cumulative errors in text line generation. In contrast, diffusion models are less susceptible to this issue.

Stylized Handwritten Text-Line Generation. We present the quantitative results on the ICDAR2013 dataset in Table 2. It is evident that models trained in the latent space, including One-DM variants and Imitating-Diff, outperform those trained in the pixel space, such as FontDiffuser and GAC-Diff. As shown in Fig. 6, we observe that models trained in the pixel space tend to generate characters with dark ink color and thick strokes, failing to capture the full range of reference styles. This phenomenon may be attributed to the excessively strong guidance of content information during diffusion training, which hinders the adequate learning of style feature extraction and fusion. Furthermore, it is reasonable that pixel-space-based methods achieve significantly higher CR and AR values, as these models tend to generate characters that closely resemble printed fonts, making them more easily recognizable.

When comparing with One-DM variants and Imitating-Diff, we observe that although they perform similarly in the FID metric, Imitating-Diff excels in style score metrics and content metrics. Combined with the visualization results in

Table 2. Compared with other methods.

	FID (↓)	GS (↓)	Style Score (↑)	CR (↑)	AR (↑)
FontDiffuser	75.61	3.66E−04	3.80	92.69	92.49
GAC-Diff	76.14	3.62E−04	7.40	92.95	92.77
OnE−DM + PE	42.30	1.90E−03	19.70	35.29	30.05
OnE−DM + LA	44.66	1.11E−03	20.10	73.19	72.33
Imitating-Diff (4 channel latent)	38.25	1.61E−04	53.40	81.51	80.75
Imitating-Diff (16 channel latent)	**33.33**	**1.12E−04**	**56.20**	89.99	89.13

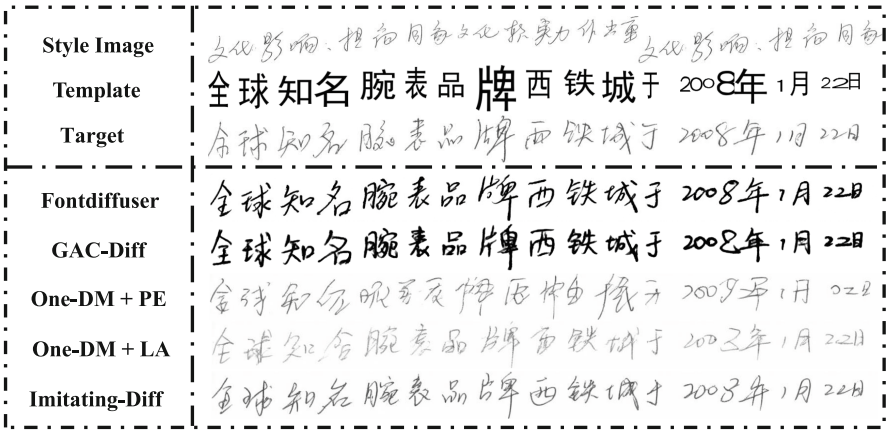


Fig. 6. Qualitative comparisons between our method with other character-based methods on handwritten text-line generation. We utilize the same template image and style reference image. Our model effectively captures the handwriting style of text lines while preserving the structural integrity of individual characters, thereby generating images of superior quality compared to other models.

Fig. 6, Imitating-Diff exhibits fewer instances of missing or redundant strokes, as well as reduced structural distortions. We attribute this improvement to the fine-tuning in pixel space, complemented by our proposed high-frequency loss. To mitigate compression-induced information loss, we replace the SD1.5 VAE [35] with the higher-capacity SD3.5 VAE [10], which increases the latent channel dimension from 4 to 16. As shown in Table 2, this enhances both structural fidelity and stylistic diversity.

Ablation Study. We use the One-DM model as our baseline. Due to its inability to generate reasonable text-line images, we omit the quantitative results of the baseline model from the Table 3. The results highlight the effectiveness of the fine-tuning loss, as performance is significantly improved in models both with and without the style-content aggregation modules. Additionally, the visual results Fig. 7 demonstrate that models trained with our proposed pixel-level loss

Table 3. Ablation Study of Proposed Modules.

	FID (↓)	GS (↓)	Style Score (↑)	CR (↑)	AR (↑)
Baseline + PE	44.38	4.85E−03	23.60	32.00	25.24
BaseLine + LA	44.90	1.70E−04	19.60	80.15	79.38
Baseline - SCF + LA	39.77	**8.61E−05**	41.70	78.05	77.12
Baseline + LA + FT	42.12	9.29E−05	37.30	80.04	80.32
Baseline - SCF + LA + FT	**38.25**	1.61E−04	**53.40**	**81.51**	**80.75**

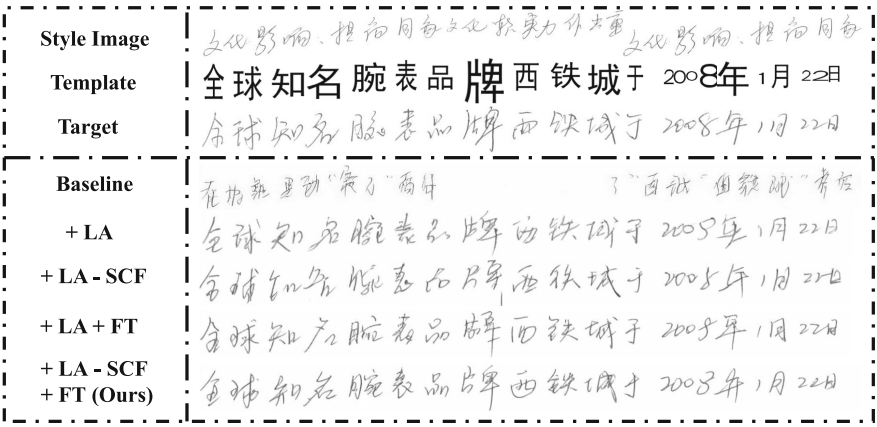


Fig. 7. Visualization of Ablation Experiments, where LA represents latent aggregation module, SCF represents style content fusion module, FT represents fine-tuning in the pixel space. We find that fine-tuning in the pixel space significantly enhances the accuracy of content structure preservation. Furthermore, the style-content fusion module proves to be unnecessary, thereby substantially reducing computational overhead.

produce characters with clearer and more defined structures. Furthermore, we observe that leaving out the style-content fusion module enhances the model's ability to mimic style characteristics. This improvement is likely due to the possibility that excessive content fusion hinders the model's ability to learn distinctive writing styles.

Linear Style Interpolation. To further investigate the latent space learned by our Imitating-Diff, we conduct linear style interpolation experiments between different writers, with the visualization results shown in Fig. 8. For the transition from writer C003 to C057, the characters evolve from cursive to neat, and the ink color intensifies from light to dark. Conversely, the transition from writer C029 to C027 demonstrates a shift in ink color from dark to light and a transformation in character structure from compact to sparse. These results reveal that the generated text-line images smoothly transition between styles, including ink



Fig. 8. Visualization of latent linear interpolation. During style transitions, the content template remains fixed while linear interpolation is applied to the extracted style features.

color and stroke thickness, while maintaining the integrity of the original textual content.

Generation with the Same Text in Different Styles. As illustrated in Fig. 9, Layout-Diff and Imitating-Diff are employed to generate text lines in 6 randomly selected styles from the test dataset, using the same content (a Chinese classical poem). We observe that writers C041, C056, and C002 exhibit larger character spacing compared to writers C005, C057, and C029. Consequently, the generated images for C041, C056, and C002 tend to be longer than those for C005, C057, and C029. Meanwhile, the ink colors are well preserved in the generated images. The results demonstrate that the proposed pipeline effectively replicates diverse styles while preserving the structural integrity in the majority

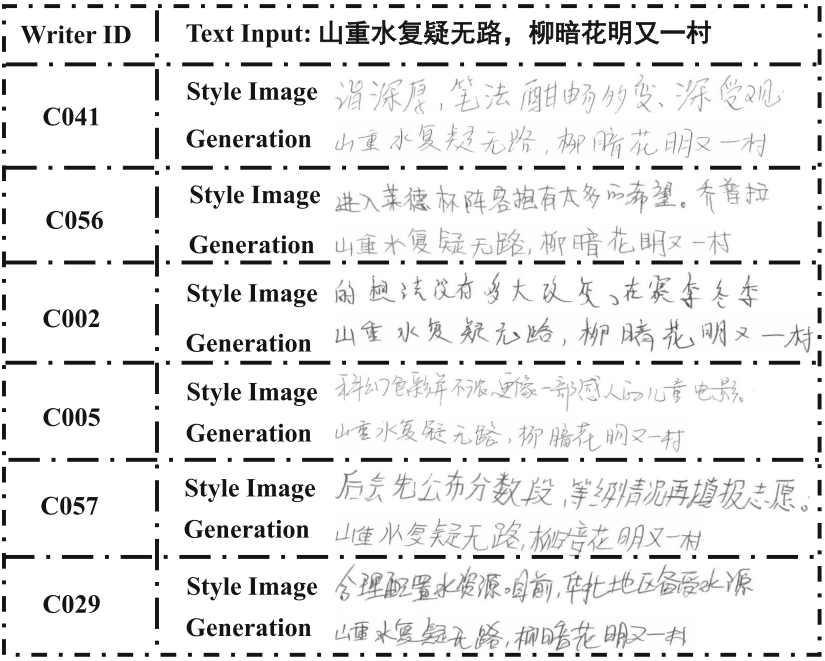


Fig. 9. The visualization of text lines generated by Layout-Diff and Imitating-Diff, based on user-provided content, demonstrates that our two-stage diffusion models effectively capture both text-line-level and character-level stylistic features.

of cases. The primary limits of performance lies in maintaining the accuracy of the glyph structure.

6 Conclusion and Future Work

In this paper, we propose a novel framework for stylized offline Chinese handwritten text-line generation that integrates glyph templates with Layout-Diff and Imitating-Diff to disentangle line- and character-level styles. Through latent aggregation and high-frequency fine-tuning, the model preserves character structure while capturing stylistic variation. Future work will explore its extension to multilingual scripts.

Acknowledgments. This work is supported by the Natural Science Foundation of China (NSFC) Grant No. U23B2029 and No. 62276258.

References

1. Aksan, E., Pece, F., Hilliges, O.: Deepwriting: making digital ink editable via deep generative modeling. In: Proceedings of the CHI Conference on Human Factors in Computing Systems, pp. 1–14 (2018)

2. Alonso, E., Moysset, B., Messina, R.: Adversarial generation of handwritten text images conditioned on sequences. In: International Conference on Document Analysis and Recognition, pp. 481–486 (2019)
3. Bao, F., Li, C., Zhu, J., Zhang, B.: Analytic-DPM: an analytic estimate of the optimal reverse variance in diffusion probabilistic models. In: International Conference on Learning Representations (2021)
4. Bhunia, A.K., Khan, S., Cholakkal, H., Anwer, R.M., Khan, F.S., Shah, M.: Handwriting transformers. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 1086–1094 (2021)
5. Dai, G., Zhang, Y., Ke, Q., Guo, Q., Huang, S.: One-DM: one-shot diffusion mimicker for handwritten text generation. In: European Conference on Computer Vision (2024)
6. Dai, G., et al.: Disentangling writer and character styles for handwriting generation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 5977–5986 (2023)
7. Das, A., Yang, Y., Hospedales, T., Xiang, T., Song, Y.Z.: ChiroDiff: modelling chirographic data with diffusion models. In: International Conference on Learning Representations (2023)
8. Dhariwal, P., Nichol, A.: Diffusion models beat GANs on image synthesis. Adv. Neural. Inf. Process. Syst. **34**, 8780–8794 (2021)
9. Ding, H., Luan, B., Gui, D., Chen, K., Huo, Q.: Improving handwritten OCR with training samples generated by glyph conditional denoising diffusion probabilistic model. In: International Conference on Document Analysis and Recognition, pp. 20–37 (2023)
10. Esser, P., et al.: Scaling rectified flow transformers for high-resolution image synthesis. In: Forty-First International Conference on Machine Learning (2024)
11. Gan, J., Wang, W.: HiGAN: handwriting imitation conditioned on arbitrary-length texts and disentangled styles. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 35, pp. 7484–7492 (2021)
12. Graves, A.: Generating sequences with recurrent neural networks. arXiv preprint arXiv:1308.0850 (2013)
13. Guan, H., et al.: Deciphering oracle bone language with diffusion models. In: Annual Meeting of the Association for Computational Linguistics (2024)
14. Gui, D., Chen, K., Ding, H., Huo, Q.: Zero-shot generation of training data with denoising diffusion probabilistic model for handwritten Chinese character recognition. In: International Conference on Document Analysis and Recognition, pp. 348–365 (2023)
15. Ha, D., Eck, D.: A neural representation of sketch drawings. In: International Conference on Learning Representations (2018)
16. Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., Hochreiter, S.: GANs trained by a two time-scale update rule converge to a local NASH equilibrium. In: Advances in Neural Information Processing Systems, vol. 30 (2017)
17. Ho, J., Jain, A., Abbeel, P.: Denoising diffusion probabilistic models 2020. In: International Conference on Learning Representations (2021)
18. Kang, L., Riba, P., Rusinol, M., Fornes, A., Villegas, M.: Content and style aware generation of text-line images for handwriting recognition. IEEE Trans. Pattern Anal. Mach. Intell. **44**(12), 8846–8860 (2021)
19. Khrulkov, V., Oseledets, I.: Geometry score: a method for comparing generative adversarial networks. In: International conference on machine learning, pp. 2621–2629 (2018)

20. Kingma, D.P., Welling, M.: Auto-encoding variational Bayes. In: International Conference on Learning Representations (2014)
21. Kong, Y., et al.: Look closer to supervise better: one-shot font generation via component-based discriminator. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 13482–13491 (2022)
22. Kotani, A., Tellex, S., Tompkin, J.: Generating handwriting via decoupled style descriptors. In: European Conference on Computer Vision, pp. 764–780 (2020)
23. Lian, Z., Zhao, B., Chen, X., Xiao, J.: Easyfont: a style learning-based system to easily build your large-scale handwriting fonts. ACM Trans. Graph. **38**(1), 1–18 (2018)
24. Lin, J.W., Hong, C.Y., Chang, R.I., Wang, Y.C., Lin, S.Y., Ho, J.M.: Complete font generation of Chinese characters in personal handwriting style. In: International Performance Computing and Communications Conference, pp. 1–5 (2015)
25. Liu, C.L., Yin, F., Wang, D.H., Wang, Q.F.: Casia online and offline Chinese handwriting databases. In: International Conference on Document Analysis and Recognition, pp. 37–41 (2011)
26. Liu, W., Liu, F., Ding, F., He, Q., Yi, Z.: XMP-font: self-supervised cross-modality pre-training for few-shot font generation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 7905–7914 (2022)
27. Lu, C., Zhou, Y., Bao, F., Chen, J., Li, C., Zhu, J.: DPM-solver: a fast ode solver for diffusion probabilistic model sampling in around 10 steps. Adv. Neural. Inf. Process. Syst. **35**, 5775–5787 (2022)
28. Lugmayr, A., Danelljan, M., Romero, A., Yu, F., Timofte, R., Van Gool, L.: Repaint: inpainting using denoising diffusion probabilistic models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 11461–11471 (2022)
29. Luhman, T., Luhman, E.: Diffusion models for handwriting generation. arXiv preprint arXiv:2011.06704 (2020)
30. Nikolaidou, K., et al.: Wordstylist: styled verbatim handwritten text generation with latent diffusion models. In: International Conference on Document Analysis and Recognition, pp. 384–401 (2023)
31. Pippi, V., Cascianelli, S., Cucchiara, R.: Handwritten text generation from visual archetypes. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 22458–22467 (2023)
32. Radford, A., Metz, L., Chintala, S.: Unsupervised representation learning with deep convolutional generative adversarial networks. In: International Conference on Learning Representations (2016)
33. Ren, M.S., Zhang, Y.M., Chen, Y.: Decoupling layout from glyph in online Chinese handwriting generation. arXiv e-prints, pp. arXiv–2410 (2024)
34. Ren, M.S., Zhang, Y.M., Wang, Q.F., Yin, F., Liu, C.L.: Diff-writer: A diffusion model-based stylized online handwritten Chinese character generator. In: International Conference on Neural Information Processing, pp. 86–100 (2023)
35. Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B.: High-resolution image synthesis with latent diffusion models (2021)
36. Sohl-Dickstein, J., Weiss, E.A., Maheswaranathan, N., Ganguli, S.: Deep unsupervised learning using nonequilibrium thermodynamics. In: Proceedings of International Conference on International Conference on Machine Learning (2015)
37. Song, J., Meng, C., Ermon, S.: Denoising diffusion implicit models. In: International Conference on Learning Representations (2020)
38. Song, Y., Ermon, S.: Generative modeling by estimating gradients of the data distribution. In: Advances in Neural Information Processing Systems, vol. 32 (2019)

39. Song, Y., Sohl-Dickstein, J., Kingma, D.P., Kumar, A., Ermon, S., Poole, B.: Score-based generative modeling through stochastic differential equations. In: International Conference on Learning Representations (2021)
40. Tang, S., Lian, Z.: Write like you: synthesizing your cursive online Chinese handwriting via metric-based meta learning. Comput. Graph. Forum **40**(2), 141–151 (2021)
41. Tang, S., Xia, Z., Lian, Z., Tang, Y., Xiao, J.: FontRNN: generating large:cale Chinese fonts via recurrent neural network. Comput. Graph. Forum **38**(7), 567–577 (2019)
42. Tolosana, R., Delgado-Santos, P., Perez-Uribe, A., Vera-Rodriguez, R., Fierrez, J., Morales, A.: DeepWritesyn: on-line handwriting synthesis via deep short-term representations. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 35, pp. 600–608 (2021)
43. Xie, Y., Chen, X., Sun, L., Lu, Y.: DG-Font: deformable generative networks for unsupervised font generation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 5130–5140 (2021)
44. Xu, P., Hospedales, T.M., Yin, Q., Song, Y.Z., Xiang, T., Wang, L.: Deep learning for free-hand sketch: a survey. IEEE Trans. Pattern Anal. Mach. Intell. **45**(1), 285–312 (2022)
45. Xu, S., Jin, T., Jiang, H., Lau, F.C.: Automatic generation of personal Chinese handwriting by capturing the characteristics of personal handwriting. In: Twenty-First IAAI Conference (2009)
46. Yang, Z., Peng, D., Kong, Y., Zhang, Y., Yao, C., Jin, L.: FontDiffuser: one-shot font generation via denoising diffusion with multi-scale content aggregation and style contrastive learning. In: Association for the Advancement of Artificial Intelligence (2024)
47. Yao, G., et al.: Geometric-aware control in diffusion model for handwritten Chinese font generation. In: International Conference on Document Analysis and Recognition, pp. 3–17 (2024)
48. Yin, F., Wang, Q.F., Zhang, X.Y., Liu, C.L.: ICDAR 2013 Chinese handwriting recognition competition. In: International Conference on Document Analysis and Recognition, pp. 1464–1470 (2013)
49. Zhang, X.Y., Yin, F., Zhang, Y.M., Liu, C.L., Bengio, Y.: Drawing and recognizing Chinese characters with recurrent neural network. IEEE Trans. Pattern Anal. Mach. Intell. **40**, 849–862 (2016)
50. Zhao, B., Tao, J., Yang, M., Tian, Z., Fan, C., Bai, Y.: Deep imitator: handwriting calligraphy imitation via deep attention networks. Patteren Recognit. **104**, 107080 (2020)
51. Zhu, J.Y., Park, T., Isola, P., Efros, A.A.: Unpaired image-to-image translation using cycle-consistent adversarial networks. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 2223–2232 (2017)
52. Zhu, Y., Li, Z., Wang, T., He, M., Yao, C.: Conditional text image generation with diffusion models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 14235–14245 (2023)